LV solo Mission ASURT

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PCB & Circuits

What is Crosstalk in PCB and How to reduce it?

Crosstalk in PCB is a phenomenon where a trace (called aggressor) creates an unwanted effect on the signal on an adjacent trace (called victim).

We can reduce crosstalk with several methods like: increasing spacing between traces, keeping a solid ground plane close to traces, and reducing the length where the traces run in parallel.

Explain the difference between Decoupling Capacitors and Bulk Capacitors and when we use them.

Decoupling capacitors and bulk capacitors are capacitors that are used to reduce the noise in a power signal by storing energy and releasing it on circuit’s demand and also providing a low impedance path to ground for high frequencies.

The main difference is their values, where to place them, and their use, decoupling capacitors have small capacitance, placed close to target circuit such as an IC, it deals with high frequency noise and sudden changes in current, while a bulk capacitor have larger capacitance, placed near the source, dealing with slower/larger changes in load current and help maintain supply voltage.

What are differential pairs in PCB? Detail the critical rules for routing them.

Differential pair is used to transmit a signal on two traces instead of one where the voltages of the transmitted signal is the difference between the two traces voltages.

This help reducing noise where the coupling between the two traces help them being exposed to the same noise which is canceled out when calculating the difference in voltages ($\mathbf{V_{diff} = (V_+ + \cancel{V_{noise}}) - (V_- + \cancel{V_{noise}})}$)

As I mentioned it's important to expose them to the same noise as possible so we need to follow some rules,

Rules for routing:

1. Keep the two traces close
2. Keep the parallel as possible
3. Match length and width

Build this circuit in your simulator and answer the following questions:

a) What is this circuit's primary function?

The circuit is a demonstration of the use of a pull-down resistor

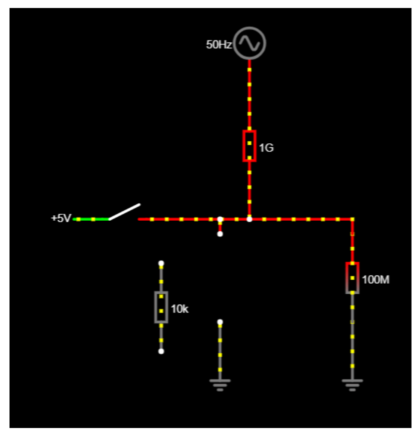
b) What do the components inside the dashed box represent? Explain in detail.

The components in the dashed box are an AC source of 50hz and a High value resistor of 1GΩ together represent a noise that will affect the output signal when the switch is open.

c) R1 (the main character)

1) What happens if you remove $R1$? (Simulate the result and explain).

The signal is affected by copying the AC voltage as there is no pull-down resistor.



2) What is an input called when a resistor like R1 is missing?

Floating or high impedance (Z)

3) How should the resistance value for R1 be selected in a real circuit?

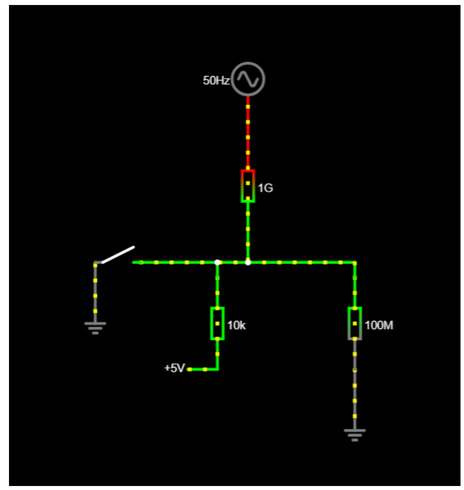
It needs to be low enough to pull down the voltage, but high enough to minimize the current it consumes, 10k is a common value as it consumes about 0.5mA in a 5V circuit.

d) What is the arrow likely pointing to, and what role does $R2$ serve? What could we call $R2$?

The arrow represents a physical boundary of a chip package,it;s role is that it illustrates how small current can leak through the IC, which can be called input resistance, or leakage resistance.

e) If our active signal voltage were 0V instead of 5V, how should the circuit be modified? What is this new circuit topology called?

If the active signal is 0V then we will need to replace the pull-down resistor with a pull-up resistor keeping the voltage at 5V (HIGH) when the switch is open.



f) What would happen if you brought your hands close to (or touched) an unshielded circuit like this?

My body will act as a human antenna injecting noise to the circuit.

Basic Embedded & Architecture

What will happen if two interrupt requests arise at the same time?

When two interrupts happen at the same time in ARM-M, the NVIC determines which one runs first by assessing their priorities as the following: the interrupt with lower group priority wins, if they are the same then, the lower subpriority wins, if the they have the same subpriority then, the one with the lower IQR runs first.

What are communication protocols in the context of embedded systems? Provide at least Three examples.

Communication protocols are a set of rules that define how MCUs communicate with each other, examples of communication protocols: UART, I2C, and CAN.

What are the differences between the following families and their use cases: STM32F4,Raspberry Pi Zero, ATmega.

STM32F4 is a family of microcontrollers based on ARM-M4 microprocessors, used in high performance industrial purposes and robotics.

ATmega is also a family of microcontrollers but it’s based on AVR architecture, have lower specs than stm32f4, it's also the microcontroller used in the famous arduino uno board, making it perfect for education, and simple projects.

On the other hand Raspberry Pi Zero is a family of single board computers(SBC) not a microcontroller, the core architecture varies with different versions, its more powerful and expensive, it can run an OS like linux making it suitable for tasks that require high computational power like computer vision for example.